Towards AI-Enabled Nuclear Astrophysics

A Preliminary Research Proposal

Lyu Shenghao[1,2]

[1]Tsinghua University, Beijing, China

[2]China Institute of Atomic Energy, Beijing, China

Sept 24, 2026



Table of Contents

1. Introduction
2. Common Machine Learning Approaches
3. Related Studies and Applications
4. AURORA: My First Research Proposal
5. Roudmap and Expected Outcomes

Introduction: Why AI?

Nuclear astrophysics is suitable for AI due to some characteristics.

- **Interdisciplinary with Knowledge Gaps**
  Relevant knowledge is distributed across nuclear physics, astrophysics, and observational literature.
- **Simulation-Based & Computationally Constrained**
  Full ucleosynthesis simulations may involve $\sim 10^3$ isotopes and up to $\sim 10^4$ reactions, coupling with 2D/3D hydrodynamics.
- **Experimentally Expensive**
  Experiments involve many coupled setup parameters, while beam time and machine access are limited and costly.
- **Uncertainty-Sensitivity Intensive**
  Make predictions and identify which uncertain inputs dominate the results should be prioritized for theory and experiments.

Introduction: ML in Nuc/Astro

Machine learning has already been widely explored in both nuclear physics and astrophysics.

Nuclear Physics

- **Nuclear theory**
  DFT emulation [1]
- **Calorimetry**
  GlueX [2]
- **Particle identification**
  DeepRICH [3]
- **Accelerator control**
  Real-time tuning [4]

Nuclear Astrophysics ⟷

Astrophysics

- **Cosmology**
  Moriwaki et al. [5]
- **Galaxy evolution**
  Fraser et al. [6]
- **Cosmic-ray analysis**
  Arimura [7]
- **Stellar abundances**
  den Hartogh et al. [8]

However, while ML has advanced both fields separately, its use in nuclear astrophysics remains very limited [9].

Common Machine Learning Approaches

AI (Artificial Intelligence) systems are largely built on ML (machine learning) methods, like deep learning. These methods are implemented through certain neural network architectures and, when trained at large scale on massive datasets, can give rise to foundation models like LLM (large language model).

Three Levels: algorithm, architecture, foundation model

Common Machine Learning Approaches

Basic Tasks: regression, classification

Learning Objectives: prediction, representation, generation, control

Modeling Paradigms: discriminative, generative

Learning Paradigms: supervised, unsupervised, reinforcement learning

Data Modalities: tabular, image, sequence, graph, multimodal

Training Objectives: supervised loss, reconstruction, contrastive learning, likelihood, reward optimization

Model Development: data, architecture, evaluation

Regression

Goal: learn continuous input–output relationships.

- **Linear Regression (LinR)**
  Fits linear relationships between inputs and outputs.
- **Kernel Ridge Regression (KRR)**
  Uses kernel mappings to model nonlinear and correlated relationships.
- **Gaussian Process (GP)**
  Models a distribution over possible functions and provides both predictions and uncertainties.

$$f(\mathbf{x}) \sim \mathcal{GP}\left(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')\right)$$

$$\mathbf{x} \longrightarrow \mu(\mathbf{x}) \pm \sigma(\mathbf{x})$$

Scientific role: prediction, interpolation, uncertainty quantification.

Classification

Goal: identify classes, structures, and unusual patterns in data.

- **Decision Tree / Random Forest**
  Tree-based classification; ensembles improve robustness.
- **Gradient Boosting**
  Sequentially combines weak predictors into stronger models.
- **Support Vector Machine (SVM)**
  Separates classes in a high-dimensional feature space.
- **K-Means / KNN**
  Groups data according to distances and similarities.
- **Isolation Forest**
  Detects rare or anomalous samples efficiently.

Scientific role: classification, clustering, pattern recognition, anomaly detection.

Neural Networks

Goal: learn complex nonlinear representations and mappings.

- **General-Purpose Networks** — MLP, DNN, ResNet
- **Structure-Aware Networks** — CNN, GNN
- **Long-Range Modeling Architectures** — RNN, LSTM, Transformer, State-Space Models

Scientific Role: Emulation

$$\boxed{\text{Expensive Simulation} \quad \mathbf{x} \to f(\mathbf{x})}$$

$$\Downarrow \text{ Training}$$

$$\boxed{\text{Fast ML Emulator} \quad \mathbf{x} \to \hat{f}(\mathbf{x})}$$

Emulators reproduce simulation input–output mappings, enabling faster parameter-space exploration, uncertainty quantification, and sensitivity analysis.

Generative Models

Goal: learn latent representations and data distributions, and generate new samples.

- **Generative Adversarial Networks (GANs)**
  Train a generator against a discriminator to produce realistic samples.
- **Diffusion Models**
  Generate data by learning to reverse a progressive noising process.
- **Variational Autoencoders (VAEs)**
  Encode high-dimensional data into a probabilistic latent space and reconstruct the input.

$$\mathbf{x} \xrightarrow{\text{Encoder}} q_\phi(\mathbf{z} \mid \mathbf{x}) \to \mathbf{z} \xrightarrow{\text{Decoder}} p_\theta(\mathbf{x} \mid \mathbf{z})$$

High-dimensional data $\to$ Latent representation $\to$ Reconstruction / Generation

Scientific role: data generation, latent-space representation.

Large Language Models

Goal: learn and process knowledge from large-scale sequential and multimodal data.

- **Large Language Model (LLM)**
  Learns statistical structures from massive text corpora.
- **Generative Pre-trained Transformer (GPT)**
  Combines large-scale pre-training with generative Transformer models.
- **Foundation Model (FM)**
  Pre-trained on large-scale, often multimodal and unlabeled datasets, then adapted to downstream domains.

$$\text{General Foundation Model} \xrightarrow{\text{Continual Pre-training}} \text{Scientific Domain Model}$$

Scientific role: knowledge integration, multi-step reasoning, agent system.

Other Algorithm Types

Complementary methods for dimensionality reduction, probabilistic inference, and interpretable discovery.

- **Principal Component Analysis (PCA)**
  Reduces high-dimensional correlated data to a smaller set of representative variables with minimal information loss.
- **Naive Bayes**
  Uses Bayesian posterior probabilities for classification and decision making.
- **Symbolic Regression (SR)**
  Searches directly for an interpretable mathematical expression describing input–output relationships.

  .....

Coscientist [10]: From Language to Experiment

Scientific Question

Can an LLM translate a natural-language synthesis goal into executable laboratory actions and correct its plan using tool feedback?

AI Principle: GPT-4 Planner with an explicit action space

- The Planner selects among **GOOGLE**, **DOCUMENTATION**, **PYTHON**, and **EXPERIMENT**; observations return to its context.
- It repaired an invalid heater–shaker call by retrieving documentation; GC–MS confirmed Suzuki–Miyaura and Sonogashira products.

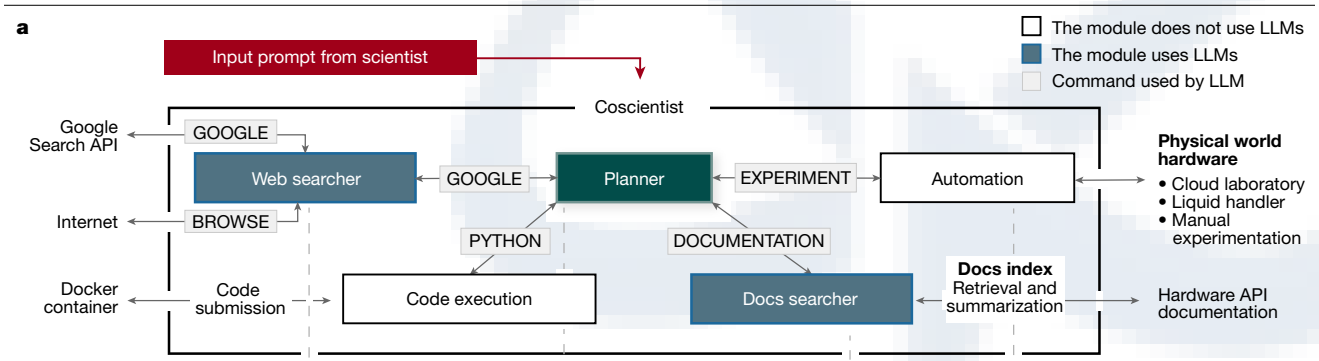


Figure 1: Coscientist tool-use architecture. Source: [10], Fig. 1a.

Robin [11]: From Data to Hypothesis

Scientific Question

Can an AI system revise a drug-repurposing hypothesis after analysing new wet-lab data? The case study is dry age-related macular degeneration.

AI Principle

- **Crow/Falcon:** concise/deep literature synthesis and candidate evaluation.
- **Finch:** Jupyter analysis of flow-cytometry and RNA-seq data; eight trajectories form a consensus.
- Human-run experiments close the hypothesis–data loop.

Results

- Ripasudil and KL001 showed in-vitro efficacy for dAMD.
- Follow-up RNA-seq implicated *ABCA1* in the ripasudil response.

Robin: Multi-Agent Architecture [11]

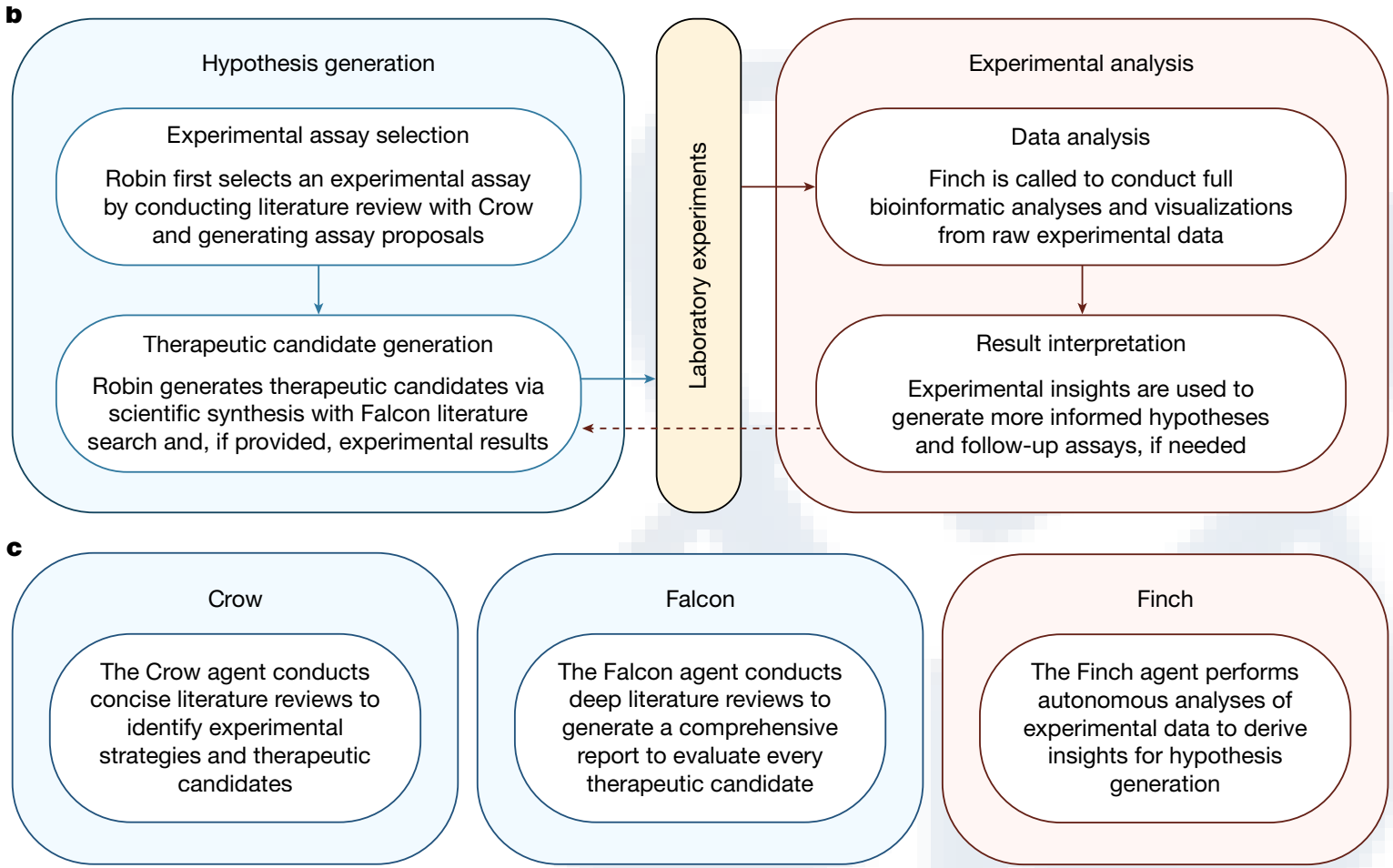


Figure 2: Literature agents and the Finch data-analysis agent form a lab-in-the-loop workflow. Source: [11], Fig. 1b,c.

XunZi [12] (1/2): Training the L-Brain

Scientific Question

How can fragmented biomedical literature and molecular data yield *functional* gene–disease hypotheses with testable mechanisms?

Technical Stack: L-Brain = XunZi-R

- **Backbone:** Mistral 7B (7.3 billion parameters).
- **Continual pretraining (CPT):** next-token learning on 24.4 million publications and 2.05 million structured biological texts.
- **Supervised fine-tuning (SFT):** 336,108 curated chain-of-thought gene–disease examples teach a Yes/No judgement and mechanistic explanation; GPT-4 drafts were manually checked.

Learning Objectives

$$\mathcal{L}_{\mathrm{CPT}} = -\sum_{t} \log p_\theta(x_t \mid x_{<t}), \qquad \mathcal{L}_{\mathrm{SFT}} = -\sum_{t} \log p_\theta(y_t \mid y_{<t}, x)$$

XunZi [12] (2/2): R-Brain and Learned Fusion

Technical Stack: R-Brain = XunZi-M

- A two-layer graph convolutional network propagates multi-omics features over PPI and Gene Ontology links.
- 2.81 million PPIs, 47,922 GO terms, and 613.2 TB of multi-omics data produce gene embedding $\mathbf{h}_i$.

Learned Fusion

$$\mathbf{z}_i = [\mathbf{h}_i \| s_i] \xrightarrow{\text{classifier}} \widehat{y}_i$$

XunZi-R contributes confidence s_i and the mechanistic explanation.

Results and Validation

- Fivefold cross-validation and reciprocal DisGeNET/CTD transfer tests.
- CHK2 and IRAK4 were nominated for Parkinson's disease; Chk2 inhibition rescued neuronal loss and motor deficits in mice.

XunZi: L-Brain/R-Brain Architecture [12]

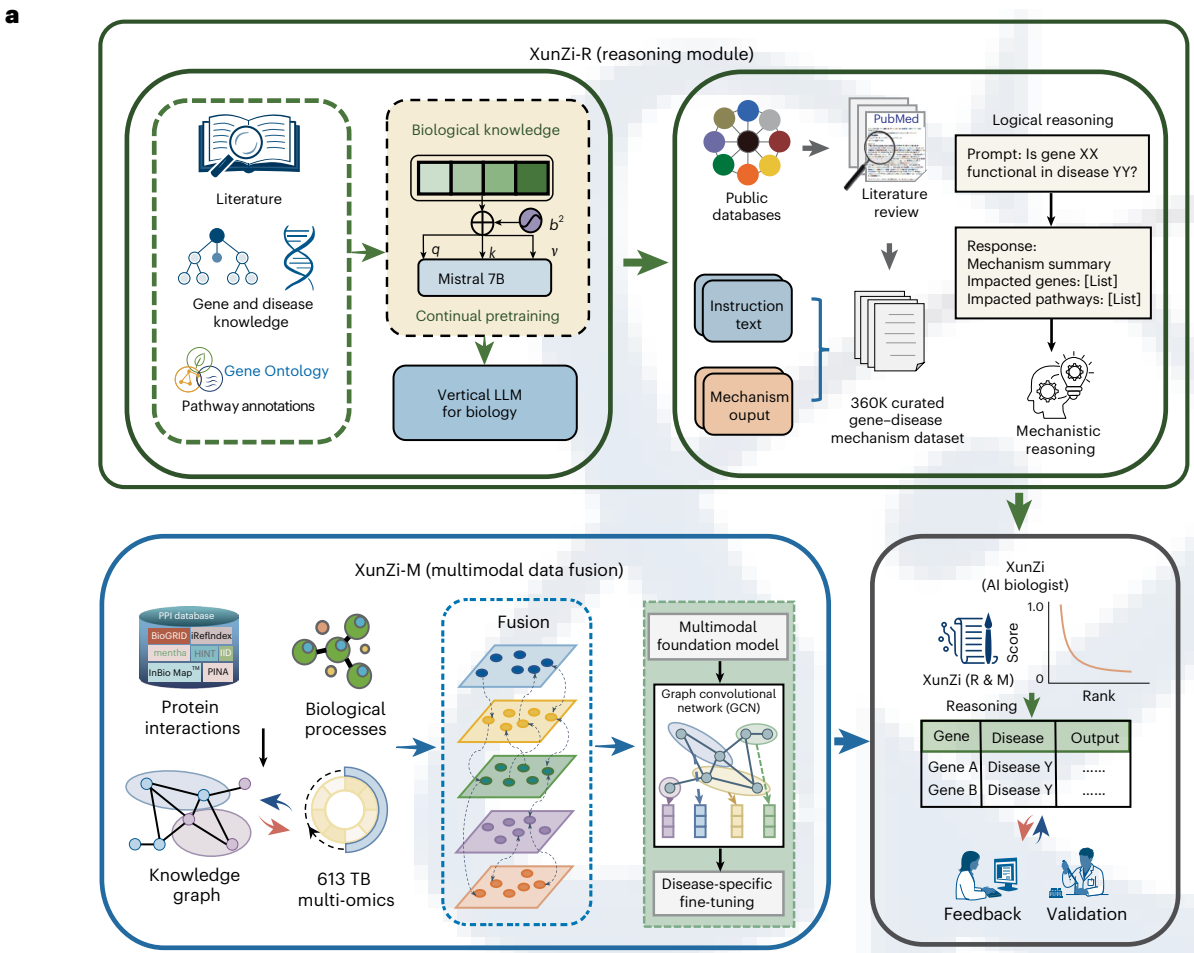
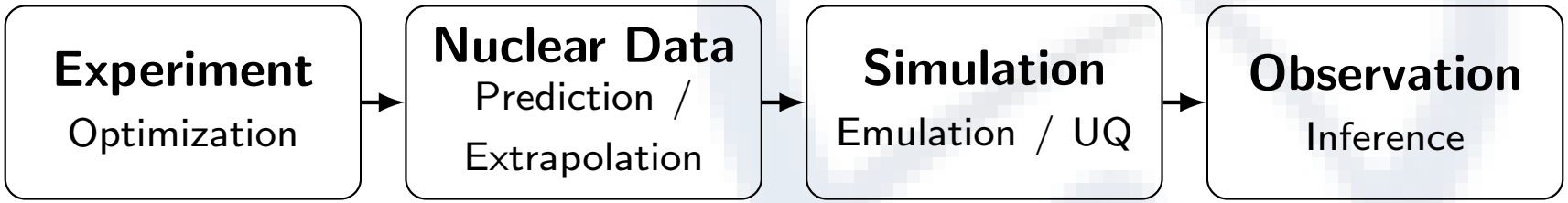


Figure 3: XunZi combines XunZi-R reasoning with XunZi-M multimodal fusion. Source: [12], Fig. 1a.

ML Across Nuclear Astrophysics

Machine learning can intervene at different stages of the nuclear-astrophysics workflow [9].



- **Prediction / Extrapolation:** learn an unknown physical quantity from known features,
- **Simulation Emulation:** approximate an expensive numerical solver,
- **Optimization:** search for optimal experimental or model parameters,
- **Inverse Inference:** infer physical parameters from observations,

Nuclear Data Extrapolation: Atomic Masses [13]

Scientific Question

How do mass predictions beyond the measured nuclei affect *r*-process abundance peaks and their uncertainty bands?

ML Problem: probabilistic nuclear-mass regression and extrapolation.

Mathematical Abstraction

$$\mathbf{f}_{Z,N} \xrightarrow{\text{MDN}} p_\phi(M_{Z,N} \mid \mathbf{f}_{Z,N})$$

$$S_n(Z,N) = M_{Z,N-1} + m_n - M_{Z,N}$$

$$\{S_n\} \xrightarrow[\text{fixed ejecta trajectory}]{\text{reaction network}} Y(A)$$

Atomic Masses: Method and Results [13]

Technical Stack

- **Training data:** 600 measured AME2020 masses ($Z \geq 10$) plus 50 HFB-32 masses.
- **Inputs:** nucleon counts, valence and parity features, and proton–neutron interaction terms.
- **Surrogate:** mixture density network with six 10-unit tanh layers; outputs mass predictions and uncertainty.
- **Physics check:** constrained loss $\mathcal{L} = \mathcal{L}_{\text{data}} + 0.3\mathcal{L}_{\text{phys}}$; propagate masses through S_n to PRISM abundances.

Results

- RMS mass error: ~ 138 keV on training data; ~ 246 keV against AME2020 for $Z \geq 10$.
- Major solar *r*-process abundance features are reproduced.
- Abundance uncertainty spans roughly one order of magnitude.

Atomic Masses: Abundance Predictions [13]

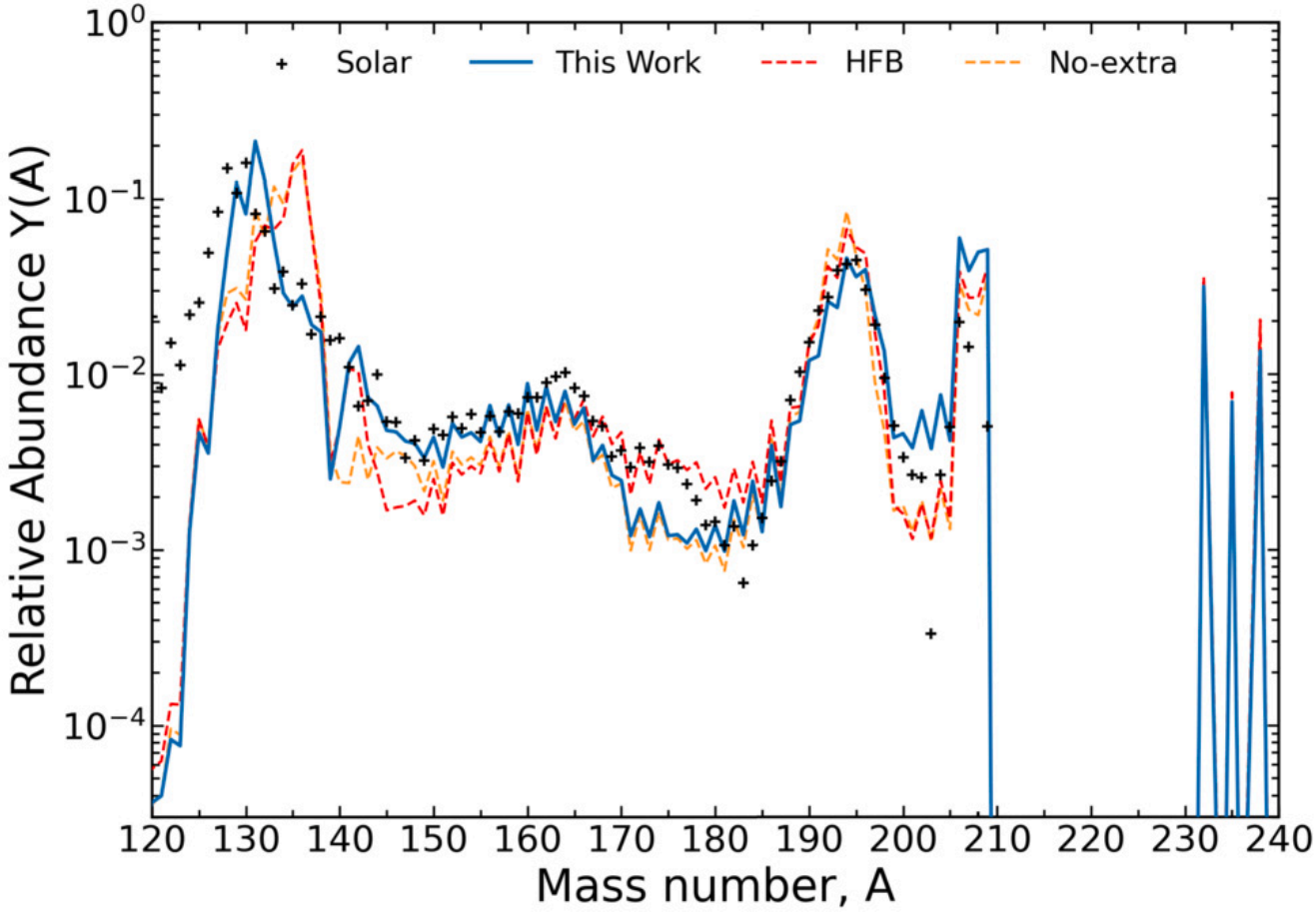


Figure 4: The predicted *r*-process abundance pattern compared with solar data. Source: [13], Fig. 4.

Atomic Masses: Uncertainty Propagation [13]

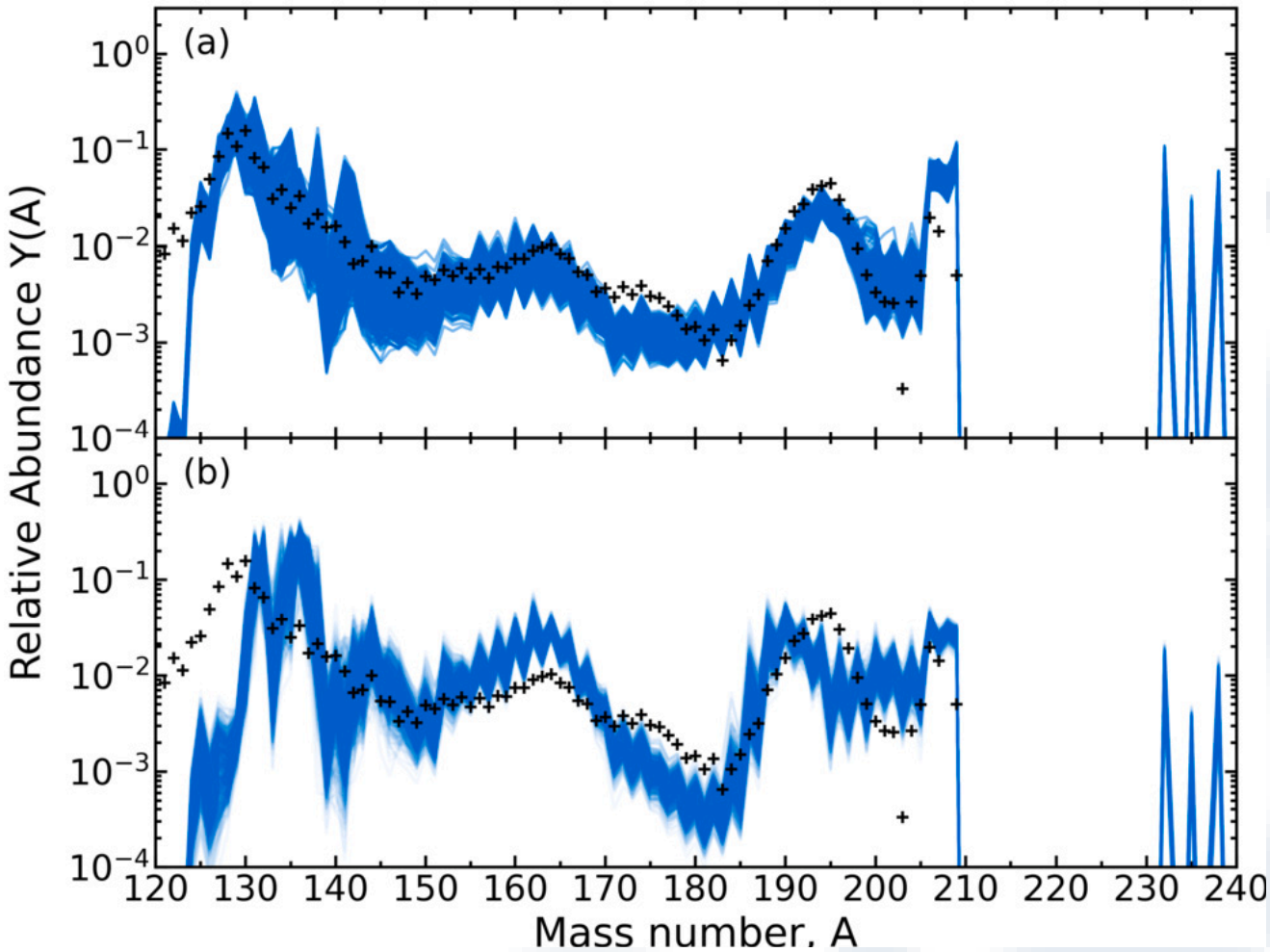


Figure 5: Mass uncertainty propagated to abundance patterns for two ejecta conditions. Source: [13], Fig. 5.

Simulation Emulation: Big Bang Nucleosynthesis [14]

Scientific Question

Can full-network BBN precision be retained while predicting primordial ^{4}He and D/H fast enough for cosmological inference?

ML Problem: code-specific, supervised abundance emulation.

Mathematical Abstraction

$$\mathbf{x} = (\Omega_b h^2, \tau_n, \Delta N_{\mathrm{eff}}, \kappa_{10})$$

$$\widehat{F}_{c,\phi}(\mathbf{x}) \approx F_c(\mathbf{x}) = (Y_{\mathrm{P}}, \mathrm{D/H})$$

c denotes PArthENoPE or AlterBBN; the latter uses equivalent code-specific input parameters.

BBNet: Method and Results [14]

Technical Stack

- **Training data:** 20,000 Latin-hypercube samples per modified full BBN solver (PArthENoPE and AlterBBN).
- **Network:** linear projection and GELU, eight-head self-attention, then residual MLP blocks with dropout.
- **Training:** AdamW; abundance error plus smoothness and D/H relative-error penalties.
- **Validation:** compare predictions with full solvers and measure downstream MCMC convergence time.

Results

- Negligible emulator error over broad parameter ranges.
- ~ 6 ms per sample; ~ 2.5 ms with batch inference.
- Up to $10^4\times$ speed-up relative to first-principles BBN calculations.

BBNet: Neural Emulator Architecture [14]

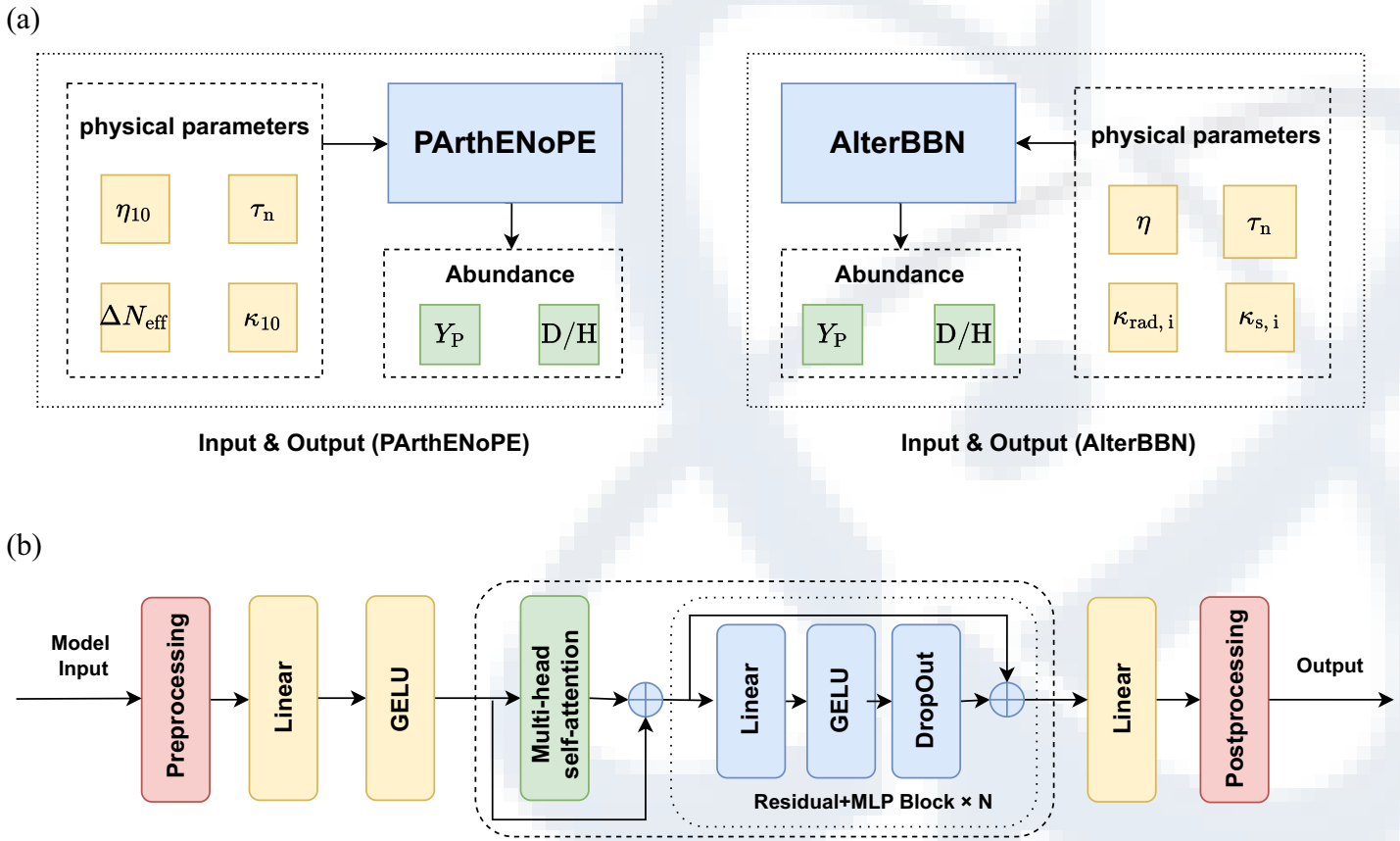


Figure 6: BBN solver inputs and the BBNet architecture. Source: [14], Fig. 2.

BBNet: Inference Speed [14]

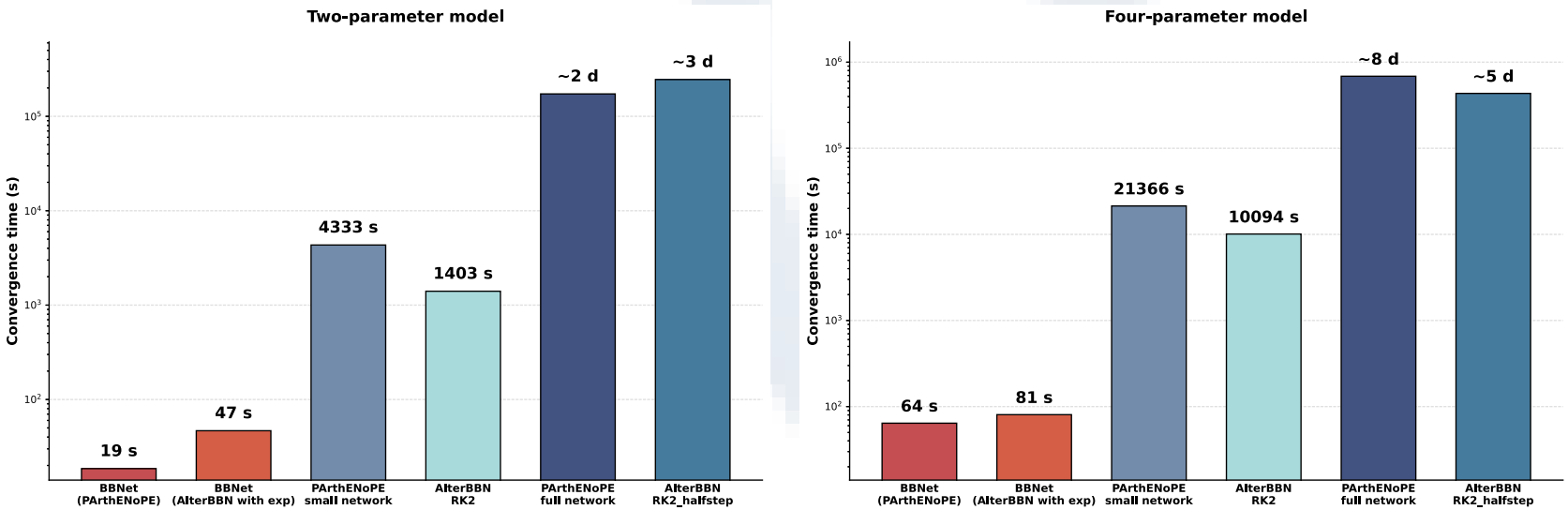


Figure 7: MCMC convergence time with BBNet and numerical BBN solvers. Source: [14], Fig. 8.

Simulation Emulation: Spontaneous Fission [15]

Scientific Question

Can a DFT surrogate reproduce fission barriers and collective inertias accurately enough to predict *r*-process fission half-lives?

ML Problem: supervised emulation of deformation-dependent DFT outputs.

Mathematical Abstraction

$$\mathbf{x} = (Z, N, Q_{20}, Q_{30}) \longmapsto \left(\widehat{V}(\mathbf{x}), \widehat{\mathcal{M}}_{ij}(\mathbf{x})\right)$$

$$(\widehat{V}, \widehat{\mathcal{M}}) \xrightarrow{\text{least-action fission path}} \widehat{t}_{1/2}^{\,\mathrm{SF}}$$

Spontaneous Fission: Method and Results [15]

Technical Stack

- **Reference data:** constrained HFB/DFT potential energy and inertia on (Q_{20}, Q_{30}) grids for 194 nuclei.
- **Surrogates:** fully connected ReLU networks with 2–7 hidden layers for the two DFT outputs.
- **Robustness:** weighted committees; separate nuclei for training, combination, and validation.
- **Physics propagation:** nudged elastic band least-action paths yield spontaneous-fission half-lives.

Results

- Potential-energy surfaces: ~ 500 keV RMSE.
- Collective inertia reproduced within roughly one order of magnitude.
- $t_{1/2}^{\mathrm{SF}}$ typically reproduced within a factor of 10^3 over more than 70 orders of magnitude.

Spontaneous Fission: DFT vs. NN [15]

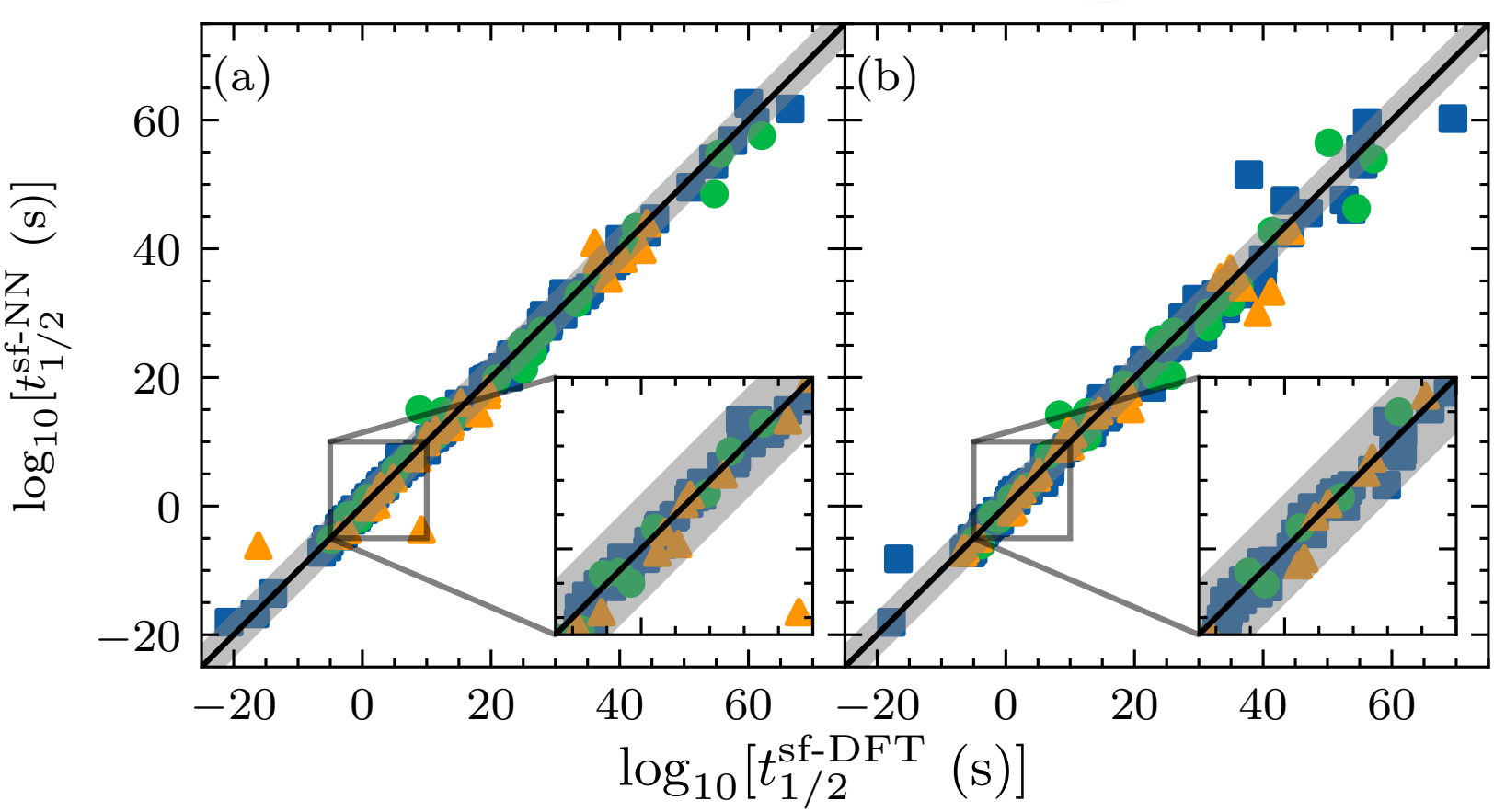


Figure 8: Spontaneous-fission half-lives from DFT and neural-network emulation. Source: [15], Fig. 6.

Simulation Emulation: *r*-Process Abundances [16]

Scientific Question

Can a neural surrogate replace repeated PRISM calculations as 424 rare-earth nuclear inputs vary, while preserving final yields?

ML Problem: high-dimensional, trajectory-specific network emulation.

Mathematical Abstraction

$$\mathbf{x} = \left(S_{n,i},\, T^{\beta}_{1/2,i}\right)_{i=1}^{212} \in \mathbb{R}^{424}$$

$$\widehat{Y}_A^{(\tau)} = f_\phi^{(\tau)}(\mathbf{x}) \approx Y_{A,\mathrm{PRISM}}^{(\tau)}(\mathbf{x}), \quad 150 \le A \le 180$$

τ: fixed merger-ejecta or hot-wind trajectory.

r-Process Emulator: Method and Results [16]

Technical Stack

- **Inputs and labels:** Sobol-sampled S_n and β-decay half-lives for 212 nuclei; PRISM final abundances.
- **Training set:** 280,000 cases for fitting and 20,000 for validation; separate merger and hot-wind models.
- **Surrogate:** strided convolutional layers followed by fully connected layers.
- **Uncertainty:** ten independently trained networks form a deep ensemble with predictive variance.

Results

- Final abundance patterns reproduced with typically $< 5\%$ deviation.
- PRISM: ~ 400 s per evaluation; emulator: ~ 0.02 s.
- Approximately 2×10^4 speed-up.

r-Process Emulator: PRISM vs. ANN [16]

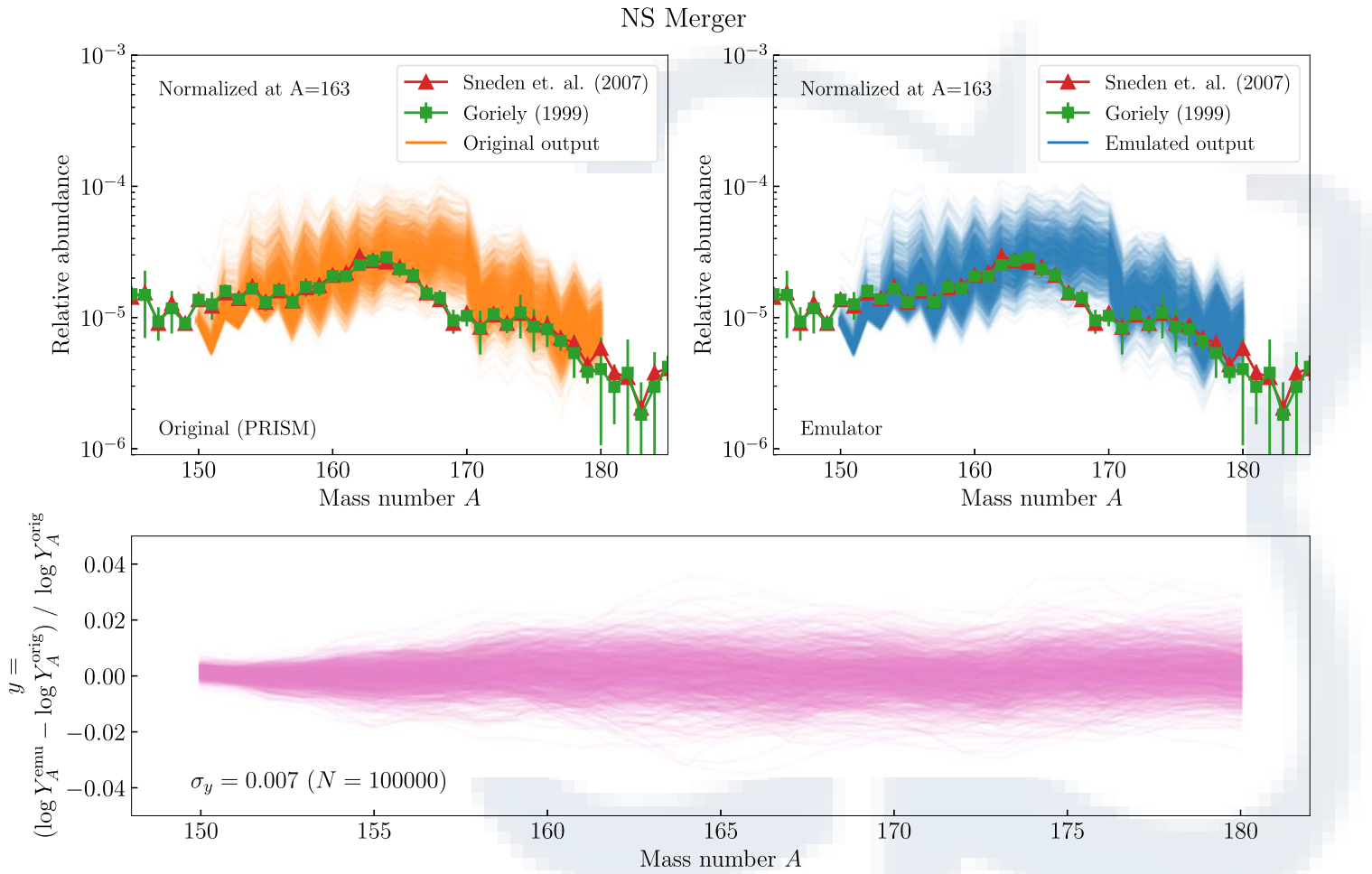


Figure 9: Rare-earth abundances from PRISM and its neural emulator. Source: [16], Fig. 3.

Exp Optimization: Online Bayesian Optimization [17]

Scientific Question

Can beam-based Bayesian optimization align SECAR and improve mass-plane optics with fewer measurements than manual tuning?

ML Problem: sample-efficient optimization of measured beam metrics.

Mathematical Abstraction

$$\mathbf{u}_s^* = \arg\min_{\mathbf{u}_s} d_{\mathrm{FP1}}(\mathbf{u}_s)$$

$$f_{\mathrm{FP2}}(\mathbf{u}_m) = \left(\frac{x + w_x - x_0}{20}\right)^4 + \left(\frac{w_y}{70}\right)^4$$

$\mathbf{u}_s$ and $\mathbf{u}_m$ are steerer and multipole settings; d_{FP1} measures the spread of four spot centers. In f_{FP2}, x_0 is the beam-axis position and w_x, w_y are spot widths [14, Eq. (3)].

Online Bayesian Optimization: Method and Results [17]

Technical Stack

- **Beam diagnostics:** FP1 spot centers under quadrupole changes; FP2 spot position and widths.
- **Surrogate:** Gaussian process with a squared-exponential kernel, updated after each beam measurement.
- **Next setting:** minimize the lower confidence bound $\mathrm{LCB}(\mathbf{u}) = \mu(\mathbf{u}) - \xi\sigma(\mathbf{u})$.
- **Online control:** GPy/GPyOpt with PyEpics; tune steerers, then multipoles, and verify beam transmission.

Results

- Beam angular deviation reduced to < 1 mrad.
- At least $3\times$ faster than manual tuning.
- Beam-spot width reduced by 32% for ^{133}Cs.

Online Bayesian Optimization at SECAR [17]

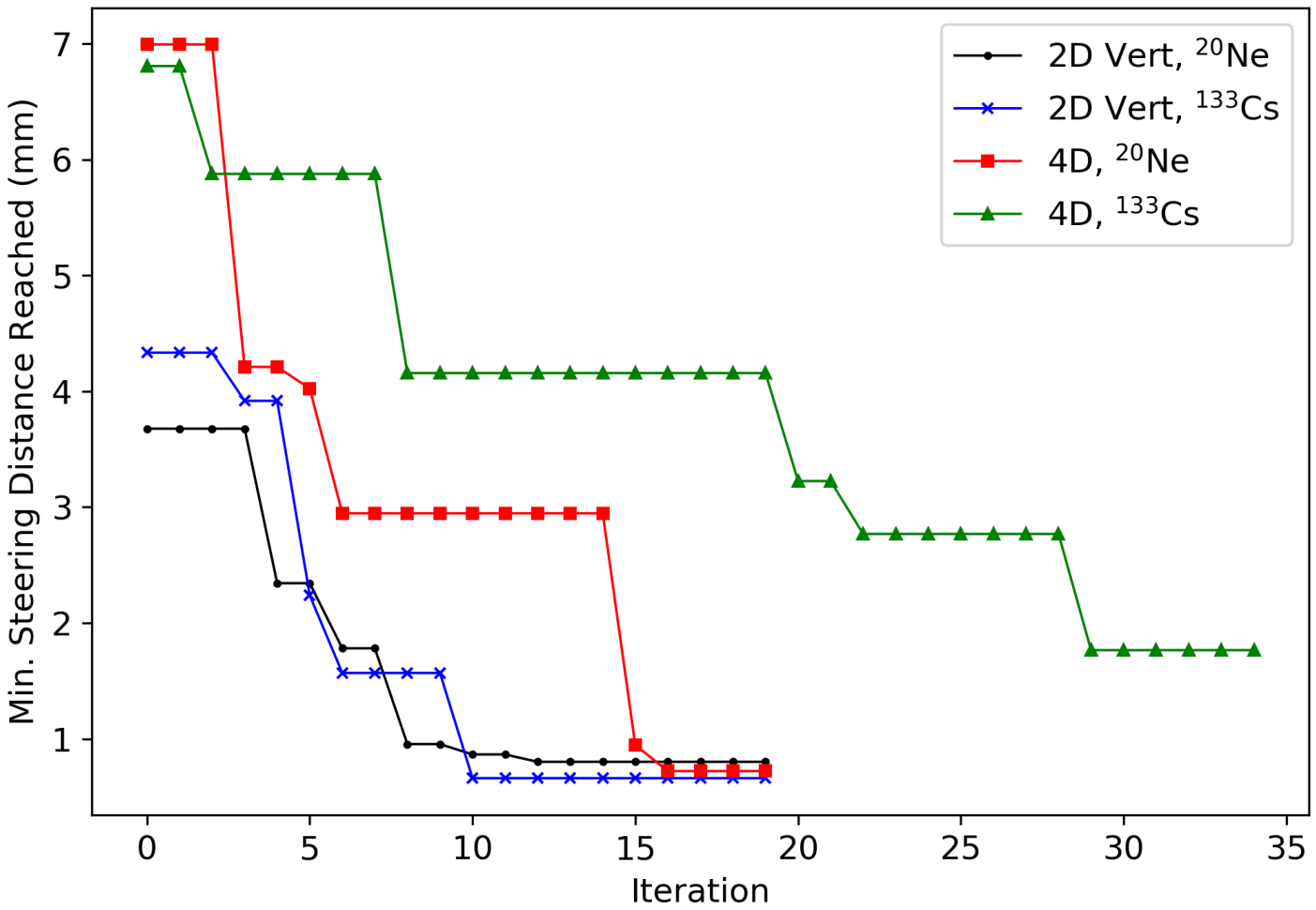


Figure 10: SECAR steering-distance convergence during online Bayesian optimization. Source: [17], Fig. 3.

Exp Optimization: SECAR (p, n) Optics [18]

Scientific Question

Which SECAR settings can separate nearly isobaric (p, n) recoils from the beam while retaining enough recoils for a cross-section measurement?

ML Problem: simulation-guided multi-objective optics optimization.

Mathematical Abstraction

$$\mathbf{u} \in \mathbb{R}^{19} \xrightarrow{\text{COSY-INFINITY}} \left(\Delta_{\mathrm{FP2}}, \Delta_{\mathrm{FP3}}, w_1, \ldots, w_K\right)$$

$$\text{maximize } \left(\Delta_{\mathrm{FP2}}, \Delta_{\mathrm{FP3}}\right), \qquad \text{minimize } \left(w_1, \ldots, w_K\right)$$

Δ: recoil–beam separation at the slits; w: beam width along the separator.

SECAR (p, n) Optics: Method and Results [18]

Technical Stack

- **Search space:** 19 field settings; increase beam–recoil separation at FP2/FP3 and limit beam widths.
- **Simulator:** COSY-INFINITY evaluates beam and recoil transport for each candidate setting.
- **Optimizer:** MOEA/D with Tchebycheff decomposition, differential evolution, and polynomial mutation.
- **Validation:** test Pareto-selected optics with beam; interpolate recoil transmission using a GP.

Results

- New ion optics enabled astrophysical (p, n) measurements with SECAR.
- Proof of principle:

$$^{58}\mathrm{Fe}(p, n)^{58}\mathrm{Co}.$$

- Measured cross section:

$$\sigma = 20.3 \pm 6.3 \text{ mb}.$$

SECAR (p, n): Optimized Ion Optics [18]

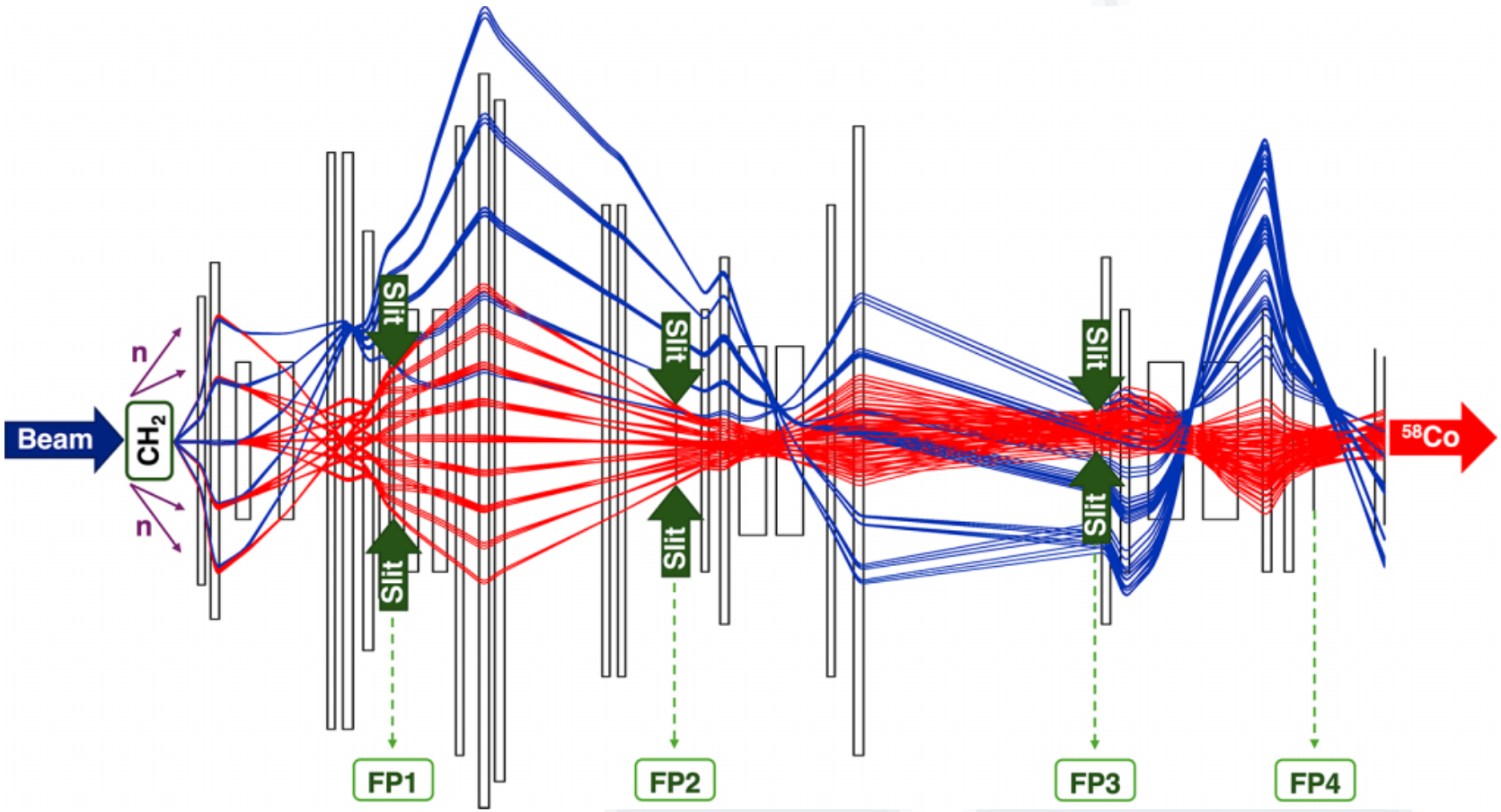


Figure 11: Optimized SECAR ion optics for (p, n) measurements. Source: [18], Fig. 1.

Astronomical Inference: Binary Neutron Stars [19]

Scientific Question

Can all 17 binary-neutron-star parameters be inferred in about one second, including sky localization before merger?

ML Problem: amortized, simulation-based posterior estimation.

Mathematical Abstraction

$$\mathbf{d} = \{\widetilde{d}_k(f)\}_{k=1}^{K}, \quad \mathbf{S}_n = \{S_{n,k}(f)\}_{k=1}^{K}$$

$$p(\boldsymbol{\theta} \mid \mathbf{d}, \mathbf{S}_n) \propto \mathcal{L}(\mathbf{d} \mid \boldsymbol{\theta}, \mathbf{S}_n)\pi(\boldsymbol{\theta})$$

$$q_\phi(\boldsymbol{\theta} \mid \mathbf{d}, \mathbf{S}_n) \approx p(\boldsymbol{\theta} \mid \mathbf{d}, \mathbf{S}_n)$$

$\boldsymbol{\theta}$ contains component masses, spins, sky location, luminosity distance, and other source parameters. The network uses an event-specific prior and importance reweighting.

DINGO-BNS: Method and Results [19]

Technical Stack

- **Training data:** simulated BNS waveforms in detector noise for neural posterior estimation.
- **Compression:** event-specific chirp-mass prior, heterodyning, multibanding, and pre-merger frequency masking.
- **Posterior model:** strain embedding plus a conditional normalizing flow trained by negative log-likelihood.
- **Correction:** importance-sample the learned posterior with the GW likelihood.

Results

- All 17 BNS parameters in ~ 1 s.
- Sky localization improved by approximately 30%.
- Supports pre-merger inference and approximately hour-long signals.

DINGO-BNS: Real-Time GW Inference [19]

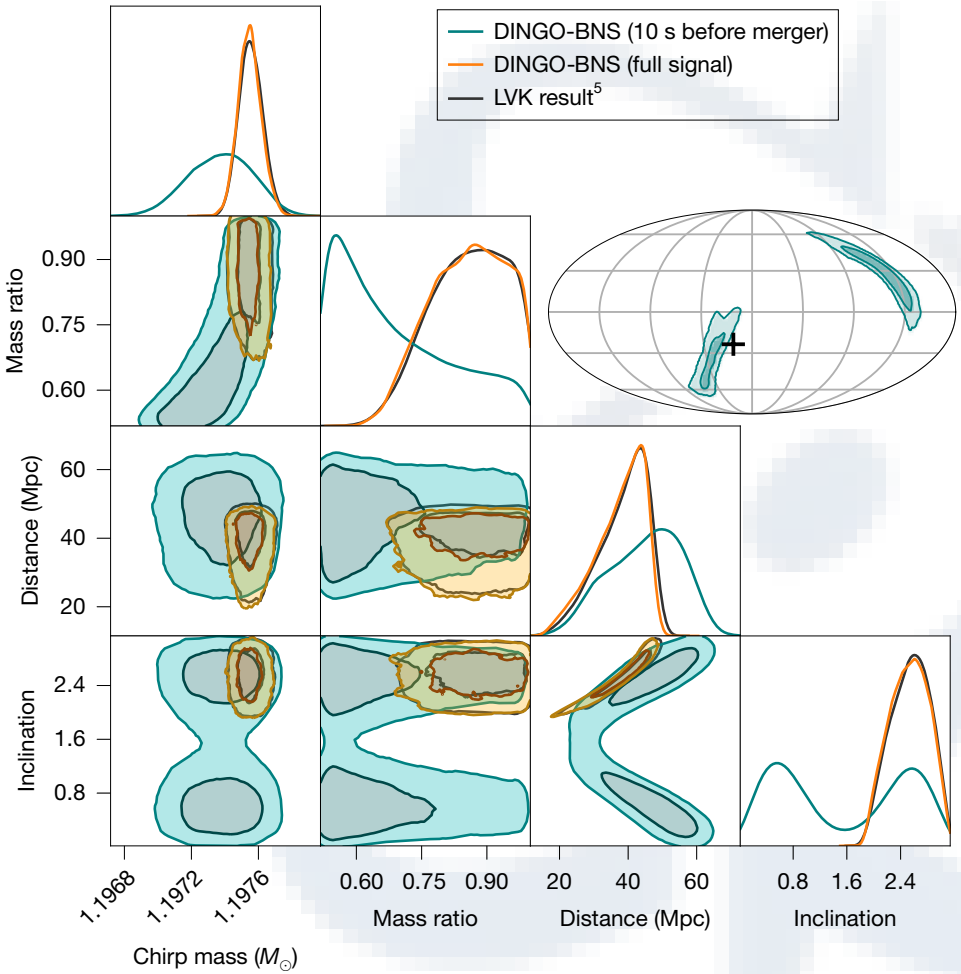


Figure 12: DINGO-BNS parameter posteriors for GW170817. Source: [19], Fig. 1.

DINGO-BNS: Pre-Merger Localization [19]

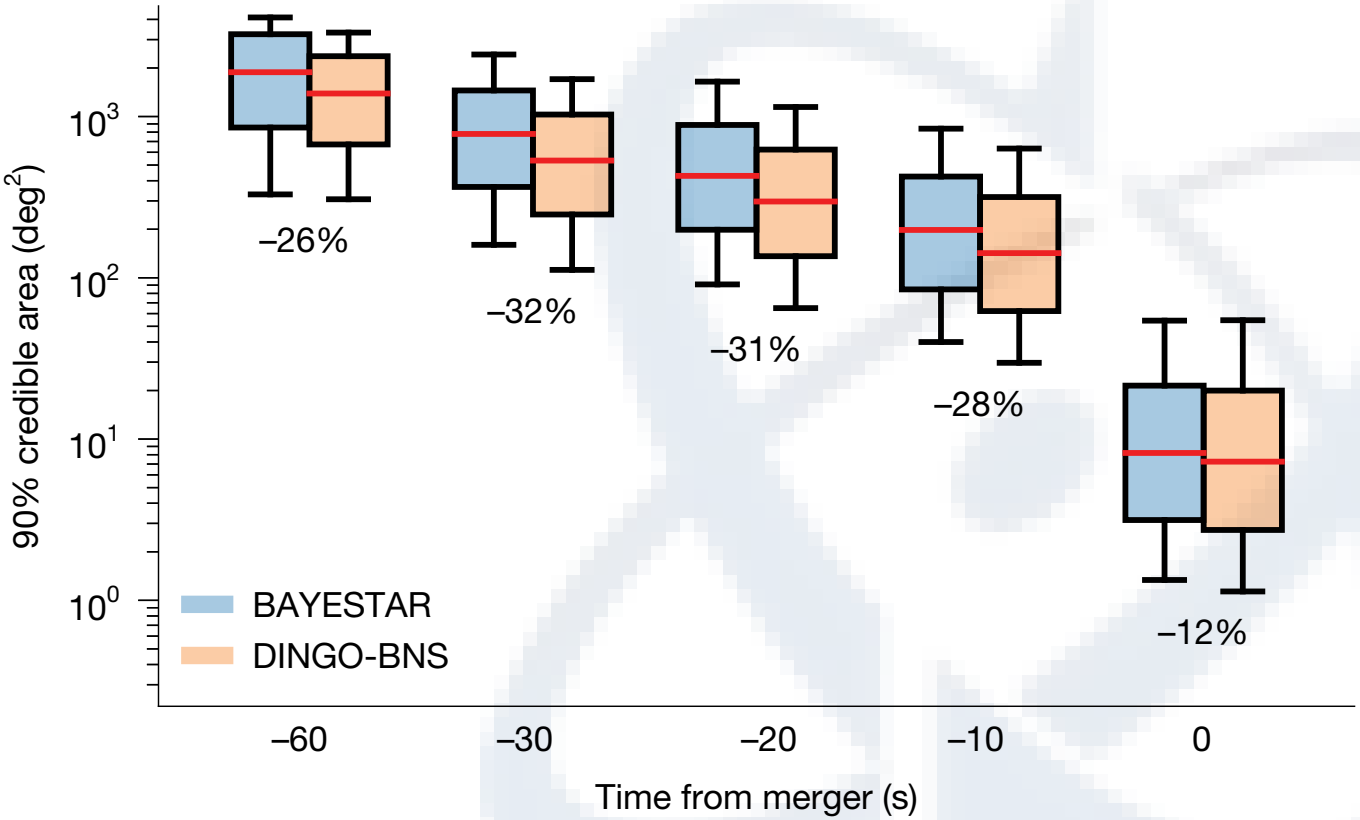


Figure 13: Pre-merger sky localization versus BAYESTAR. Source: [19], Fig. 2b.

AURORA

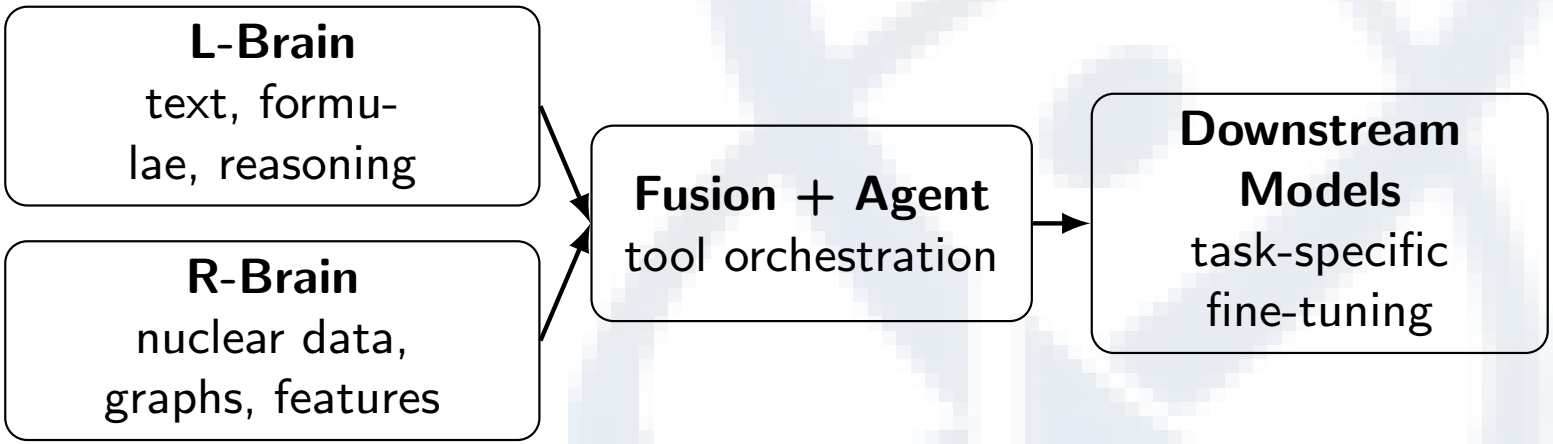
AI for Understanding the Reaction Of NucleaR Matter in Astrophysics

AURORA: AI System + LLM for Nuclear Astrophysics

Goal

Build a scalable, generalizable, and interpretable nuclear-astrophysics Co-Scientist that integrates NLP and multimodal learning for:

- knowledge question answering; mathematical derivation and code/tool use;
- experiment optimization and control; ultimately, autonomous discovery.



Development strategy: reproduce the XunZi-style two-brain paradigm in nuclear astrophysics, control each implementation detail, then redesign modules in response to observed failures. The upstream L/R-Brains are shared across multiple downstream tasks; additional modules can be introduced when required.

AURORA: Parallel Workstreams

Data line — core contribution

- Build high-quality data whose distribution differs from general LLM corpora.
- Prepare L-Brain text/formula data and R-Brain multimodal nuclear data.
- Prepare downstream fine-tuning sets and independent evaluation sets.
- Clean, normalize, deduplicate, preprocess, and audit all data.

Model line

- Train one NLP model, one multimodal model, and one or more downstream models.
- Select architectures, loss functions, and training strategies.
- Run controlled baselines and ablation studies for every module.
- Evaluate capabilities after each training stage.

Science line

- Select the discovery task with domain experts.
- Define physical constraints and independent validation.
- Expand into a dedicated line when the science task is fixed.

Human contribution: domain specialists may be required for data annotation, manual review of generated reasoning traces, and scientific sign-off.

AURORA: L-Brain Data and Training

Nuclear-Astrophys-Corpus

- Textbooks, papers, and formula-rich nuclear-astrophysics material.
- Target scale: 5B+ tokens of cleaned, structured, and deduplicated data.
- Automated PDF/LaTeX parsing, cleaning, and synthesis pipeline.
- Human audit of source quality; LLM-generated examples supplement, but do not replace, curated data.

Training and evaluation

- Start from a strong open-weight NLP model.
- **CPT:** internalize domain language, equations, and concepts.
- **SFT:** learn professional QA, formula derivation, and CoT reasoning.
- Evaluate Accuracy and CoT quality on AstroMLab and a dedicated QA benchmark.
- Target: outperform GPT-4o and Qwen2.5-72B on the domain benchmarks.

raw sources $\rightarrow$ parse $\rightarrow$ clean/deduplicate $\rightarrow$ expert audit $\rightarrow$ CPT $\rightarrow$ SFT

AURORA: R-Brain Data and Training

Scientific operator to be learned *following the task map in Ref. [9]*

$$\mathcal{R}_\phi : \underbrace{\{\mathbf{Y}_0,\ \lambda_k(T),\ T(t), \rho(t), Y_e(t)\}}_{\text{initial composition, rates, trajectory}} \longmapsto \underbrace{\{\mathbf{Y}(t),\ F_k(t),\ \dot{E}_{\mathrm{nuc}}(t)\}}_{\text{abundances, reaction flows, energy generation}} .$$

Astrophys-Data: aligned scientific views

- Curate nuclear properties and rate curves with units, uncertainties, validity ranges, and provenance.
- Generate simulation pairs across representative environments and physical conditions; reserve independent codes/runs for testing.
- Align tables and curves with reaction-flow graphs, nuclide-chart sequences, and hydrodynamic tracer histories.
- Graph nodes encode (N, Z) and abundances; edges encode reaction identity, rate, and flux. Time-dependent views retain causal order.

Representation, prediction, and trust

- Use graph encoders for flow topology and sequence/image encoders for abundance maps and trajectories.
- Train an emulator/operator head for abundance histories, final yields, and energy generation, with calibrated uncertainty.
- Share the latent scientific state with L-Brain while retaining task-specific prediction heads.
- Enforce positivity, conservation, and reaction-network residuals; flag inputs outside the training support instead of trusting extrapolation.
- Evaluate fidelity, speed, calibration, and physical violations on held-out simulations, environments, and nuclear-data perturbations.

heterogeneous scientific data → shared physical state → operator prediction + uncertainty

AURORA: Fusion, Agents, and Scientific Validation

Division of labor

L-Brain formulates questions, retrieves evidence, derives equations, and plans tools. **R-Brain** evaluates numerical states, graphs, trajectories, and uncertainty. The Agent executes and audits the joint workflow.

Downstream task ladder *derived from Ref. [9] and the preceding ML studies*

Task family	Scientific question	Auditable output
Nuclear-data completion	Which rates/cross sections are missing, anomalous, or most uncertain?	Curves with uncertainty, anomaly flags, validity range
Flow discovery	Which waiting points, bottlenecks, cycles, and pathways control the yields?	Ranked reactions/nuclides and interpretable flow subgraphs
Fast simulation	How do rates and trajectories map to abundance histories, final yields, and energy?	Calibrated emulator and reduced network with speed/fidelity report
Inference and decision	Which inputs dominate uncertainty, what should be measured next, and which conditions explain observations?	Sensitivity ranking, active-learning proposal, posterior over physical conditions

Agent loop: question $\rightarrow$ hypothesis $\rightarrow$ model/simulator call $\rightarrow$ uncertainty and physical checks $\rightarrow$ revised hypothesis.

Validation: attach evidence, calibrated uncertainty, conservation and positivity checks, domain-of-validity status, and independent MESA/SkyNet or reaction-network reproduction to every scientific claim. Use symbolic regression only for relations stable across held-out conditions.

Roadmap: Months 1–8

Period	Main task	Planned milestone
Months 1–2	Domain knowledge foundation	Build the nuclear-astrophysics corpus and establish L-Brain.
Months 3–4	Scientific data foundation	Organize multimodal scientific data and establish R-Brain.
Month 5	System integration	Connect both brains through the Agent and select the downstream study.
Months 6–7	Scientific study	Execute the selected task and complete independent physical validation.
Month 8	Consolidation and release	Complete the paper, benchmark, and reproducible research package.

Knowledge → Scientific data → Integrated system → Validated result → Release

Expected Outcomes and Acceptance Criteria

Deliverable	Concrete output	Acceptance criterion
Nuclear-Astrophys-Corpus	5B+ token text/formula corpus and automated cleaning/synthesis pipeline	Deduplicated, structured, source-audited; independent evaluation split
L-Brain	CPT+SFT domain LLM	QA Accuracy and CoT quality above GPT-4o/Qwen2.5-72B targets
Astrophys-Data	Nuclear data plus aligned flow graphs, abundance sequences, and trajectory–simulation pairs	Common schema; units, uncertainty, validity range, and provenance retained
R-Brain	Shared representation and operator model for curves, graphs, and time-dependent states	Held-out simulator fidelity, calibrated uncertainty, physical constraints, and OOD detection
Downstream Agents	Flow analysis, emulation, UQ/sensitivity, and inference workflows	Task-specific gains in accuracy or cost, with traceable evidence and uncertainty
Science result	Ranked control points, experimental priority, inferred conditions, or an interpretable relation	Conservation/domain checks and independent reaction-network or MESA/SkyNet replication
Research package	Paper, Astrophys-Bench, reviewed code, SOP, Docker image	Baselines and ablations complete; reproducible public release

Budget: CNY 50,000

Category	CNY	Share	Purpose and basis
Cloud compute	30,000	60%	L-Brain CPT: 8×A100/H800, about 280 node-hours (20,000); R-Brain alignment: 4×A100, about 120 node-hours (7,000); Step-4 baselines and ablations (3,000).
Local hardware	12,500	25%	One RTX 4090/5090D-class GPU or local compute node for Months 1–8: data-pipeline debugging, PDF/LaTeX parsing, RAG vector database, and local code debugging.
Data and APIs	5,500	11%	GPT-4o/Claude extraction, synthesis, and audit of nuclear-astrophysics CoT data (3,500); LLM-as-a-Judge scoring and cross-checking of derivations/code (2,000).
Publication & open source	2,000	4%	Open-access/publication support and maintenance of GitHub/Hugging Face releases.
Total	**50,000**	**100%**	Covers Steps 1–5 and all stated deliverables.

Risk Register and Contingencies (1/2)

Risk	Impact	Level	Prevention / response
Insufficient samples or selection bias	High	High	Add simulated data and augmentation; if necessary, change the task or widen scope to nuclear physics $\cup$ astrophysics.
Low reliability/validity of key measurements	High	Medium	Run a pilot study and explicit sanity checks before model-scale experiments.
Sensitive-data leakage or privacy violation	Extreme	Medium	Prefer public datasets; otherwise suspend publication or convert to restricted research.
Loss of partner/platform data access	High	High	Verify citation and access rights in advance and monitor permission changes.

Risk Register and Contingencies (2/2)

Risk	Impact	Level	Prevention / response
Equipment failure or blocked software/algorithms	Medium	Medium	Use a unified container environment; preserve all logs and checkpoints.
Loss of key personnel or delayed funding	Medium	Low	Maintain detailed handover documents and SOPs; prepare a cautious contingency funding plan.
No physically novel AI discovery	Medium	High	Step-4 fallback: reproduce a known conclusion or deliver the first nuclear-astrophysics foundation-model benchmark.
Hidden software bug invalidates results	High	High	Require independent reproduction and cross-validation before scientific claims.

Decision rule: scientific novelty is a stretch goal; physical validity, reproducibility, and a useful benchmark remain mandatory outputs.

Thank You

Questions and Discussion

References

[1] R. Lasseri, D. Regnier, J. Ebran, and A. Penon, "Taming Nuclear Complexity with a Committee of Multilayer Neural Networks," *Phys. Rev. Lett.* **124**, 162502 (2020).

[2] R. Barsotti and M. R. Shepherd, "Using machine learning to separate hadronic and electromagnetic interactions in the GlueX forward calorimeter," *JINST* **15**, P05021 (2020).

[3] C. Fanelli and J. Pomponi, "DeepRICH: Learning deeply Cherenkov detectors," *Mach. Learn.: Sci. Technol.* **1**, 015010 (2020).

[4] A. Scheinker et al., "Model-independent tuning for maximizing free electron laser pulse energy," *Phys. Rev. Accel. Beams* **22**, 082802 (2019).

References

[5] K. Moriwaki, T. Nishimichi, and N. Yoshida, "Machine learning for observational cosmology," *Rep. Prog. Phys.* **86**, 076901 (2023).

[6] T. S. Fraser, R. Tojeiro, and H. G. Chittenden, "Applying unsupervised learning to resolve evolutionary histories and explore the galaxy-halo connection in IllustrisTNG," *MNRAS* **522**, 5758–5774 (2023).

[7] R. Arimura, "A machine learning approach for mass composition analysis with TALE-SD data," *EPJ Web Conf.* **283**, 02011 (2023).

[8] J. W. den Hartogh et al., "Barium stars as tracers of s-process nucleosynthesis in AGB stars. II. Using machine learning techniques on 169 stars," *Astron. Astrophys.* **672**, A143 (2023).

References

[9] M. S. Smith and D. Lu, "Machine learning opportunities for nucleosynthesis studies," *Front. Astron. Space Sci.* **11**, 1494439 (2024).

[10] D. A. Boiko et al., "Autonomous chemical research with large language models," *Nature* **624**, 570–578 (2023).

[11] A. E. Ghareeb et al., "A multi-agent system for automating scientific discovery," *Nature* **655**, 497–505 (2026).

[12] X. Huang et al., "XunZi, an AI biologist, reveals disease-modifying targets," *Nature Biomedical Engineering*, doi:10.1038/s41551-026-01769-6 (2026).

References

[13] M. Li, T. M. Sprouse, B. S. Meyer, and M. R. Mumpower, "Atomic masses with machine learning for the astrophysical r process," *Phys. Lett. B* **848**, 138385 (2024).

[14] F. Zhang, H. Diao, B. Li, J. Meyers, and P. R. Shapiro, "Accurate neural network emulator for primordial light element abundances," *Phys. Rev. D* **114**, 043535 (2026).

[15] D. Lay, E. Flynn, S. A. Giuliani, W. Nazarewicz, and L. Neufcourt, "Neural Network Emulation of Spontaneous Fission," arXiv:2310.01608 (2024).

[16] Y. Saito, I. Dillmann, R. Krücken, M. R. Mumpower, and R. Surman, "Emulation of the calculations of final r-process abundance patterns with a neural network," *J. Phys. G: Nucl. Part. Phys.* **52**, 055201 (2025).

References

[17] S. A. Miskovich et al., "Online Bayesian Optimization for the SECAR Recoil Mass Separator," arXiv:2202.01978 (2022).

[18] P. Tsintari et al., "Machine learning enabled measurements of astrophysical (p, n) reactions with the SECAR recoil separator," *Phys. Rev. Research* **7**, 013074 (2025).

[19] M. Dax et al., "Real-time inference for binary neutron star mergers using machine learning," *Nature* **639**, 49–53 (2025).

[20] J. Lei, "FUSION," GitHub repository, `github.com/jinleiphys/FUSION`.